

AYE

TECH HUB

ELECTRICAL SAFETY
ENGINEERING HANDBOOK

Electrical Safety Handbook

LOTO Procedures · Arc Flash Protection · NEC/IEC Standards
PPE Selection · Safe Work Practices · Incident Prevention

10

CHAPTERS

50+

STANDARDS

20+

PROCEDURES

FREE

ACCESS



DANGER — ARC FLASH HAZARD

Unauthorized persons must not approach energized equipment
without proper PPE and authorized work permit

~~NEPA 705 · ISO 69479 · OSHA 29 CFR 1910.333~~

Awet G. Nway

Founder, AYE Tech Hub | Electrical Engineering & Safety Expert

FREE

OPEN ACCESS

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Fundamentals of Electrical Safety

Electrical safety is not optional — it is the fundamental obligation of every engineer, technician, and worker who interacts with electrical systems. Each year, electrical hazards cause thousands of injuries, hundreds of fatalities, and billions in property damage globally. Understanding the physics of electrical hazards is the first step toward preventing incidents.

Why Electricity is Dangerous

Electric current passing through the human body causes injury through three mechanisms:

- Electrocution (cardiac arrest):** As little as 50–100mA AC at 60Hz through the chest can cause ventricular fibrillation. This is why the "let-go threshold" (16mA AC) is a critical design parameter.
- Electrical burns:** High current causes resistive heating of body tissues (I^2R). Internal burns may not be visible externally but are often more severe than surface burns.
- Arc flash injuries:** Electrical arcs can reach 35,000°F (19,400°C) — hotter than the sun's surface — causing severe burns, blast overpressure, and shrapnel injuries.

Current Effects on the Human Body

Current (AC, 60Hz)	Duration	Physiological Effect	Hazard Level
0.5 – 2 mA	Any	Perception threshold — tingling sensation	Nuisance
2 – 10 mA	Any	Pain, muscle contraction begins	Low Risk
10 – 16 mA	Any	"Let-go threshold" — cannot release grip	DANGER
16 – 50 mA	Any	Severe pain, respiratory difficulty	HIGH RISK
50 – 100 mA	1–3 sec	Ventricular fibrillation (cardiac arrest)	POTENTIALLY FATAL
100 – 200 mA	< 1 sec	Ventricular fibrillation, lung damage	FATAL
> 200 mA	Any	Heart clamps, severe burns — may survive with CPR	CRITICAL
> 1 A	Any	Severe tissue destruction, cardiac damage	ALMOST CERTAINLY FATAL

Critical Safety Insight

It is CURRENT, not voltage, that kills — but voltage determines current through body resistance.
Dry skin resistance: 100,000 to 600,000 Ω. Wet/broken skin: as low as 1,000 Ω.
At 240V with wet hands: $I = 240/1000 = 240\text{mA}$ — easily FATAL.
Even 50V DC can be fatal under the wrong conditions (wet skin, heart condition).

Voltage Classification (IEC 60038)

Classification	AC Voltage Range	DC Voltage Range	Typical Applications
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Extra Low Voltage (ELV)	< 50 V AC	< 120 V DC	Control circuits, instrumentation, SELV systems
Low Voltage (LV)	50 V – 1,000 V AC	120 V – 1,500 V DC	Building wiring, motor controls, industrial panels
Medium Voltage (MV)	1 kV – 35 kV	1.5 kV – 75 kV	Distribution networks, substations, wind turbines
High Voltage (HV)	35 kV – 230 kV	>75 kV	Transmission lines, large power plants
Extra High Voltage (EHV)	>230 kV	—	National grid interconnections

Hazard Identification & Risk Assessment

Systematic hazard identification is the foundation of any electrical safety management system. Before any work on or near electrical equipment, a formal risk assessment must identify all potential hazards, evaluate their risk level, and establish control measures using the Hierarchy of Controls.

The Hierarchy of Controls

Control Level	Description	Effectiveness
ELIMINATION	Remove the hazard entirely — most effective	■■■■■
SUBSTITUTION	Replace with a less hazardous alternative	■■■■
ENGINEERING CONTROLS	Guards, interlocks, insulation, barriers	■■■
ADMINISTRATIVE CONTROLS	Procedures, training, permits, LOTO	■■
PPE	Last resort — protect the worker	■

Risk Assessment Matrix

Risk Level = Severity × Likelihood. All electrical work must be evaluated before proceeding:

Severity \ Likelihood	Rare (1)	Unlikely (2)	Possible (3)	Likely (4)	Almost Certain (5)
Catastrophic (5)	Medium	High	Critical	Critical	Critical
Major (4)	Low	Medium	High	Critical	Critical
Moderate (3)	Low	Medium	Medium	High	Critical
Minor (2)	Negligible	Low	Medium	Medium	High
Negligible (1)	Negligible	Negligible	Low	Low	Medium

Job Safety Analysis (JSA) Steps

- Step 1 — List job tasks:** Break the job into discrete sequential steps.
- Step 2 — Identify hazards:** For each step, identify potential electrical, mechanical, and environmental hazards.
- Step 3 — Assess risk level:** Apply the risk matrix to determine the risk level for each hazard.
- Step 4 — Establish controls:** Select controls from the Hierarchy for each identified hazard.
- Step 5 — Assign responsibility:** Designate responsible persons for each control measure.
- Step 6 — Review and sign:** All workers performing the job must review and sign the JSA before starting.

■ Always Remember

No electrical work is so urgent that it cannot be done safely.

If in doubt — STOP. Consult a qualified engineer before proceeding.

Complacency is the greatest safety hazard — most incidents happen to experienced workers.

CHAPTER 3

Energy Isolation · OSHA 1910.147 · IEC 60204

Lockout/Tagout (LOTO) Procedures

Lockout/Tagout (LOTO) is the most critical electrical safety procedure in industrial environments. It ensures that hazardous energy is isolated and cannot be re-energized while workers are performing maintenance or service. OSHA 29 CFR 1910.147 (The Control of Hazardous Energy) mandates LOTO programs for all facilities where workers may be exposed to hazardous energy.



LOTO Sequence — OSHA 29 CFR 1910.147 / IEC 60204-1
Figure 3.1 — 6-Step LOTO Sequence per OSHA 29 CFR 1910.147

LOTO Step-by-Step Procedure

- **1. Notify Affected Personnel**
Inform all affected workers that equipment will be locked out. Identify all energy sources.
- **2. Shut Down Equipment**
Follow the normal stopping procedure. Engage all stop buttons and controls.
- **3. Identify All Energy Sources**
Electrical (main + all secondary feeds), pneumatic, hydraulic, thermal, gravitational, chemical.
- **4. Isolate Energy Sources**
Open all disconnect switches, close all isolation valves, block all gravity hazards.
- **5. Apply Lockout Devices**
Each worker applies THEIR OWN padlock to each isolation point. Tags alone are NOT lockout.
- **6. Release/Restrain Stored Energy**
Bleed pneumatic/hydraulic systems. Discharge capacitors. Block elevated loads.
- **7. Verify Zero Energy State**
Test with calibrated meter: confirm ZERO voltage at all work points before touching.

LOTO Equipment Requirements

Equipment	Standard Requirement	Notes
Padlock	Individually keyed — each worker has their own	Never share keys during LOTO
Hasp	Multi-lock hasp for group LOTO with multiple workers	All locks must be applied

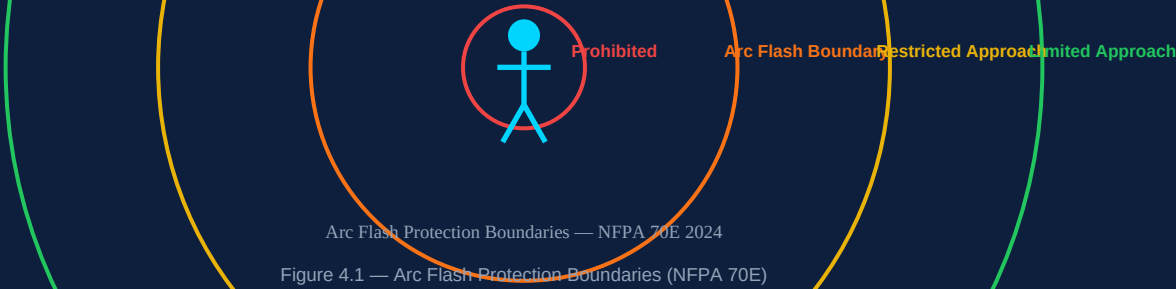
Tag	Durable, standardized warning tag attached to each lockout point	Tag supplements lock — never replaces it
Lockout stations	Wall-mounted station with all LOTO devices	Keep fully stocked at all times
Voltage tester	CAT III or CAT IV rated for voltage present	Must be tested before and after use
LOTO log	Record each LOTO event: date, worker, equipment, energy sources	Required for OSHA compliance

CHAPTER 4

Incident Energy · Boundaries · PPE Selection

Arc Flash Protection

An **arc flash** is an explosive electrical discharge through ionized air. Arc flash incidents release enormous amounts of thermal energy in milliseconds, with plasma temperatures reaching 35,000°F (19,400°C). A properly engineered arc flash protection program — per NFPA 70E and IEEE 1584 — determines the incident energy at working distance and prescribes the appropriate PPE.



Arc Flash Boundary Definitions

Boundary	Definition	Who May Cross	Action Required
Limited Approach Boundary	Shock hazard — energized conductor nearby	Qualified & unqualified persons (with escort)	Shock protection PPE required for unqualified
Restricted Approach Boundary	Increased shock risk — accidental contact possible	Qualified persons only	Shock protection PPE + insulated tools mandatory
Arc Flash Boundary	Incident energy at this distance = 1.2 cal/cm ² (bare skin burn)	Qualified persons with PPE	Arc-rated PPE at minimum HRC 1 required
Prohibited Approach	Same risk as direct contact with live part	Qualified — emergency only	Maximum PPE level; engineering controls preferred



Arc Flash PPE Categories — NFPA 70E Table 130.5(G)
Figure 4.2 — Arc Flash PPE Categories (NFPA 70E Table 130.5(G))

■ Arc Flash Study Requirements

An Arc Flash Hazard Analysis (AFHA) per IEEE 1584 must be performed for all electrical systems.

Labels must be placed on all electrical equipment showing: Incident Energy (cal/cm^2), Boundary distances, PPE category, Arc flash boundary.

Studies must be updated every 5 years or when significant system changes occur.

Equipment labels are MANDATORY per NFPA 70E and OSHA requirements.

CHAPTER 5

Arc-Rated · Insulated · Standard Requirements

Personal Protective Equipment (PPE)

PPE is the last line of defense in the electrical safety hierarchy. It must be selected based on the specific hazard level determined by the risk assessment or arc flash study — never chosen arbitrarily. Incorrect PPE can create a false sense of security or fail to provide adequate protection.

Electrical PPE Selection Guide

PPE Item	Standard	Voltage Range	Notes
Class 00 Insulated Gloves	ASTM D120	500V AC max	Thin — must fit under leather protectors
Class 0 Insulated Gloves	ASTM D120	1,000V AC max	Most common for LV work
Class 2 Insulated Gloves	ASTM D120	17,000V AC max	Medium voltage switching
Class 4 Insulated Gloves	ASTM D120	36,000V AC max	High voltage substation work
Arc-Rated FR Clothing (HRC1)	NFPA 70E	Any (≤ 4 cal/cm ²)	FR shirt + pants; safety glasses + face shield
Arc Flash Suit (HRC4)	NFPA 70E	Any (≤ 40 cal/cm ²)	Multi-layer coverall; arc-rated hood; balaclava
Dielectric Safety Boots	ASTM F2413	Class 75 (18,000V)	Must be inspected before each use
Safety Glasses/Goggles	ANSI Z87.1	All work	Mandatory at all times near electrical equipment
Hard Hat — Class E	ANSI Z89.1	20,000V proof tested	Class E required for electrical environments
Arc-Rated Face Shield	NFPA 70E	Cal/cm ² rated	Must match or exceed HRC category being worked in

■ PPE Inspection Rules

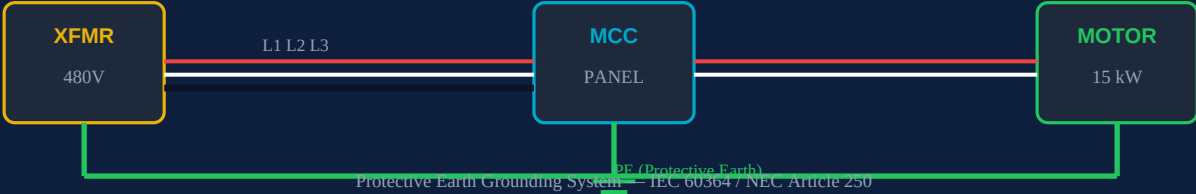
- Inspect ALL PPE before every use — even new equipment can be defective.
- Insulated gloves must be air-tested and visually inspected for cuts, tears, or swelling.
- Arc-rated clothing must be free of hydrocarbon contamination (oil, grease) — they fuel fires.
- Never wear polyester, nylon, or synthetic fibres near arc flash hazards — they melt to skin.
- Tag and remove any PPE that shows damage. Never use damaged electrical PPE.

CHAPTER 6

NEC Article 250 · IEC 60364 · TN/TT/IT Systems

Electrical Grounding & Bonding

Grounding (earthing) and bonding are critical safety systems that protect personnel from electric shock and equipment from damage due to fault currents, lightning, and static electricity. A properly designed grounding system ensures that fault current returns safely to the source — tripping protective devices — rather than flowing through personnel.



Grounding System Types (IEC 60364)

System	Description	Neutral Grounding	Common Application
TN-S	Separate PE and N conductors throughout	Solidly grounded at source	Modern industrial facilities, new buildings
TN-C	Combined conductor (PEN (N+PE combined))	Solidly grounded at source	Older European installations, not recommended
TN-C-S	TN-C to distribution board, TN-S from board	Solidly grounded at source	Residential/commercial distributions
TT	Transformer grounded; loads use separate earth electrode	Solidly grounded, separate earth at load	Rural areas, temporary supplies
IT	No direct earth connection at source; impedance earthed	Isolated (ungrounded) or impedance grounded	Hospitals, critical process plants

Grounding vs Bonding

- **Grounding:** Connecting electrical system conductors (neutral, equipment frames) to earth (ground electrode). Purpose: fault protection, lightning protection, voltage stabilization.
- **Bonding:** Connecting metal components together to ensure they are at the same electrical potential. Purpose: prevent dangerous voltage differences between metal parts that personnel might touch simultaneously.
- **Equipment Ground Conductor (EGC/PE):** The green or bare wire that connects equipment frames to the supply earth. Must carry full fault current for sufficient time to trip the breaker.

■ Ground Resistance Requirements

Single electrode system: NEC 250.56 requires $< 25 \Omega$. Otherwise, supplement with additional electrodes.

Critical facilities (data centers, hospitals): Often design for $< 5 \Omega$ or $< 1 \Omega$.

Lightning protection ground: $< 10 \Omega$ per IEC 62305.

Test ground resistance annually with a 3-point fall-of-potential test (Megger ground tester).

Never rely on conduit, water pipes, or structural steel as the only ground path.

Safe Work Practices & Permit Systems

Safe work practices for electrical work are mandated by NFPA 70E, OSHA 29 CFR 1910.333, and IEC 60204-1. The fundamental principle is: **de-energize before working**. When de-energizing is not feasible, a formal Energized Electrical Work Permit must be completed and approved by authorized management before work begins.

Energized Electrical Work Permit Requirements

Per NFPA 70E 130.2(B), an Energized Electrical Work Permit must include:

- **Description of circuit/equipment:** Voltage, amperage, location, panel ID.
- **Justification for energized work:** Why cannot the circuit be de-energized? (Production critical, continuous monitoring required, etc.)
- **Shock hazard analysis results:** Approach boundaries (limited, restricted).
- **Arc flash hazard analysis results:** Incident energy level, AFB distance, PPE category.
- **PPE to be worn:** Specific items required for shock and arc flash protection.
- **Work methods:** Specific safe work practices, tools to be used.
- **Responsible parties:** Signatures of performing worker, supervisor, and safety officer.
- **Emergency plan:** Who to call, where AED is located, nearest hospital.

Minimum Safe Approach Distances (NEC/NFPA 70E)

System Voltage (Phase-Phase)	Limited Boundary	Restricted Boundary	Prohibited Boundary
< 50 V (ELV)	Not specified	Not specified	Not specified
50 V – 300 V LV	3.0 m (10 ft)	1.0 m (3 ft 6 in)	Avoid contact
300 V – 750 V	3.0 m (10 ft)	1.0 m (3 ft 6 in)	Avoid contact
750 V – 2 kV	3.0 m (10 ft)	1.0 m (3 ft 6 in)	Avoid contact
2 kV – 15 kV MV	3.0 m (10 ft)	1.7 m (5 ft 6 in)	19 cm (7.5 in)
15 kV – 36 kV MV	3.0 m (10 ft)	2.7 m (9 ft)	24 cm (9.5 in)
36 kV – 46 kV	3.0 m (10 ft)	2.7 m (9 ft)	27 cm (10.5 in)

CHAPTER 8

Electric Shock · Burns · Arc Flash Response

Emergency Response & First Aid

Rapid, correct emergency response saves lives. All workers in electrical environments must be trained in emergency response procedures, including CPR, AED operation, and the correct response to electrical burns and arc flash injuries.

Electric Shock Emergency Response

→ **1. DO NOT TOUCH THE VICTIM**

You could also be electrocuted. Assess the scene first.

→ **2. DISCONNECT THE POWER**

Immediately de-energize the circuit. Use isolation switch — NOT by pulling the victim.

→ **3. CALL EMERGENCY SERVICES**

Call immediately — 911 (US), 999 (UK), 112 (EU), or site emergency number.

→ **4. CHECK RESPONSIVENESS**

Once power is off, approach safely. Check for pulse and breathing.

→ **5. BEGIN CPR IF NEEDED**

If no pulse: 30 chest compressions + 2 rescue breaths. Repeat until AED or EMS arrives.

→ **6. USE AED**

Apply AED as soon as available. Follow AED voice instructions precisely.

Electrical Burn First Aid

- **Do NOT remove burned clothing** that is stuck to skin — this causes further damage.
- Cool minor burns with cool (not cold) running water for 10–20 minutes.
- **Do NOT apply ice, butter, or creams** to electrical burns.
- Cover loosely with a sterile non-stick bandage. Do not wrap tightly.
- **All electrical burns require hospital evaluation** — internal tissue damage is often worse than the surface injury suggests.
- Watch for signs of internal injuries: abdominal pain, confusion, cardiac arrhythmia.

+ Site Emergency Preparedness Requirements

AED must be accessible within 3–5 minutes walking distance in all electrical work areas.

Emergency contact numbers must be posted at every electrical room entrance.

First Aid trained personnel must be available on every work shift.

Emergency evacuation routes from all electrical rooms must be clearly marked.

Conduct emergency response drills (including simulated electrical incidents) at least annually.

CHAPTER 9

NFPA 70E · IEC 60364 · OSHA · NEC

Standards, Regulations & Compliance

Compliance with electrical safety standards is legally required and ethically obligatory. Understanding the regulatory landscape ensures that safety programs are comprehensive, current, and defensible during regulatory inspections.

Standard / Regulation	Issuer	Scope	Key Requirements
NFPA 70E (2024 ed.)	NFPA (USA)	Electrical safety in the workplace	Arc flash PPE, LOTO, energized work permits, training requirements
OSHA 29 CFR 1910.269	US Dept of Labor	Electric power generation/distribution	Approach distances, PPE, qualified worker definitions
OSHA 29 CFR 1910.333	US Dept of Labor	General electrical safety	Working on energized equipment procedures
NEC (NFPA 70) 2023	NFPA (USA)	Electrical installation	Wiring methods, grounding, breaker sizing, installation safety
IEC 60364 Series	IEC (International)	Low voltage electrical installations	System design, protection, wiring, testing
IEC 60204-1	IEC	Safety of machinery	Electrical equipment on machines — emergency stop, LOTO interface
IEC 60479	IEC	Effects of current on human body	Physiological data for circuit protection design
EN 50110	CENELEC (EU)	Operation of electrical installations	EU live working procedures, approach distances
AS/NZS 3000	Standards Australia	Wiring rules (Australia/NZ)	Wiring standards, earthing requirements

Qualified vs Unqualified Person Definitions

NFPA 70E and OSHA define "qualified person" as someone who has been trained in and demonstrated proficiency in construction and operation of electrical equipment and the associated hazards. Key distinctions:

- Qualified Person:** Trained in electrical safety. May cross the Restricted Approach Boundary with appropriate PPE.
- Unqualified Person:** Not trained in electrical safety. May not cross the Limited Approach Boundary without a qualified escort.
- Authorized Person:** Has received specific authorization (usually in writing) to perform a specific task on specific equipment.
- Qualified Electrician:** Licensed professional (journeyman/master) qualified to perform electrical installation and maintenance work.

■ Compliance Program Requirements

Written electrical safety program required per NFPA 70E 110.1.

Training records must be maintained for all workers — per OSHA and NFPA 70E.

Arc flash analysis must be updated every 5 years or after system changes.

LOTO program must be documented, reviewed annually, and audited periodically.

All electrical equipment must be labeled with arc flash and voltage hazard information.

Incident Investigation & Prevention

Every electrical incident — whether it results in injury, equipment damage, or was a "near miss" — must be thoroughly investigated. Root cause analysis (RCA) identifies the systemic failures that allowed the incident to occur, enabling corrective actions that prevent recurrence. A near miss today is a fatality prevented.

Incident Investigation Process

Phase	Activity	Responsible Party	Timeframe
Immediate Response	Stabilize scene, provide first aid, secure area, notify management	Supervisor + First Aider	Minutes
Preserve Evidence	Photograph scene, collect samples, tag equipment (do not repair yet)	Safety Officer	Hours
Witness Interviews	Interview all witnesses independently, document statements	Safety Officer + HR	24–48 hours
Root Cause Analysis	Apply 5-Why analysis, Fishbone diagram, or Fault Tree Analysis	Safety Team + Engineering	48–72 hours
Corrective Actions	Develop, assign, and implement corrective and preventive actions (CAPA)	Management Engineering +	Days–Weeks
Close-Out & Lessons Learned	Share findings across site/organization; update procedures	Safety Manager	30–90 days

Root Cause Categories (Common Electrical Incidents)

- Inadequate Training

Worker performed task without proper safety training. Fix: Mandatory certified training program.
- Procedure Non-Compliance

Known procedures existed but were bypassed. Fix: Management accountability, behavioral safety program.
- Missing/Inadequate Procedures

No written procedure existed for the task. Fix: Develop procedures for all electrical tasks.
- Equipment Failure

Protective device (breaker, fuse) failed to operate. Fix: Preventive maintenance program, testing schedule.
- Design Deficiency

Original design lacked adequate protection. Fix: Engineering review, upgrade protection.

★ Building a Safety Culture

Encourage near miss reporting — reward reporting, never punish reporters.

Leadership must visibly participate in safety activities (walkdowns, toolbox talks).

"If you see something, say something" — every worker is responsible for safety.

Measure safety leading indicators (training hours, near misses reported) not just lagging indicators (injuries).

QUICK REFERENCE — VOLTAGE, PPE & DISTANCES

PPE Selection by Voltage Level

Voltage Range	Shock PPE	Minimum Eye Protection	Arc Flash PPE
< 50 V (ELV)	Not required (take care)	Safety glasses	Not typically required
50 V – 600 V (LV)	Class 0 insulated gloves + leather protectors	Safety glasses + face shield	Min HRC 1 (4 cal/cm²)
600 V – 5 kV (MV)	Class 1 insulated gloves	Safety glasses + face shield	Per arc flash study
5 kV – 15 kV (MV)	Class 2 insulated gloves	Full arc flash hood	Per arc flash study, min HRC 3
15 kV – 36 kV (MV)	Class 3 insulated gloves	Full arc flash hood + suit	Per arc flash study, min HRC 4
>36 kV (HV)	Class 4 insulated gloves	Full arc flash hood + suit	Per arc flash study — specialist required

Emergency Contacts & Response Numbers

■ **Site-Specific Emergency Numbers (Fill in for your facility)**

Site Emergency: _____
Plant Manager: _____
Safety Officer: _____
Local Fire/EMS: 911 (US) / 999 (UK) / 112 (EU)
Nearest Hospital: _____
Poison Control: 1-800-222-1222 (US)

Body Part Touch Voltage Limits (IEC 60479)

These conventional touch voltage limits are used for protective device design:

Condition	AC Touch Voltage Limit	DC Touch Voltage Limit	Basis
Dry skin, normal environment	50 V AC rms	120 V DC	IEC 60479-1
Wet/damp conditions	25 V AC rms	60 V DC	IEC 60479-1
Submerged/swimming pool	12 V AC rms	30 V DC	IEC 60479-1
Medical / patient areas	10 mV – 100 mV	—	IEC 60601 (special environments)

REFERENCES & REGULATORY RESOURCES

Key Standards & Regulations

- NFPA 70E (2024) — Standard for Electrical Safety in the Workplace.
- NFPA 70 — National Electrical Code (NEC), 2023 edition.
- OSHA 29 CFR 1910.147 — Control of Hazardous Energy (LOTO).
- OSHA 29 CFR 1910.333 — Selection and use of work practices (Electrical).
- IEC 60364 — Low-voltage electrical installations.
- IEC 60204-1 — Safety of machinery: Electrical equipment of machines.
- IEC 60479-1 — Effects of current on human beings and livestock.
- IEEE 1584-2018 — Guide for Performing Arc Flash Hazard Calculations.
- ISO 45001:2018 — Occupational health and safety management systems.

Learning & Training Resources

Resource	Website	Description
Resource	Website	Description
AYE Tech Hub	ayetechub.com	Free electrical safety courses, guides, and engineering resources
NFPA	nfpa.org	Access NFPA 70E standard, arc flash training, free safety resources
OSHA eTools	osha.gov/etools	Free online OSHA compliance tools, incident reporting
IEEE Xplore	ieeexplore.ieee.org	IEEE 1584 arc flash standard and technical papers
ISA	isa.org	Electrical safety certifications and industrial automation standards

AYE Tech Hub — Engineering the Future

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